Title: Apple Conveyer System.

Objectives:

- To understand and apply basic PLC ladder programming in Rockwell CCW.
- To implement a count up instruction for apple counting.

System Description:

The automated system consists of two main conveyors and two sensors:

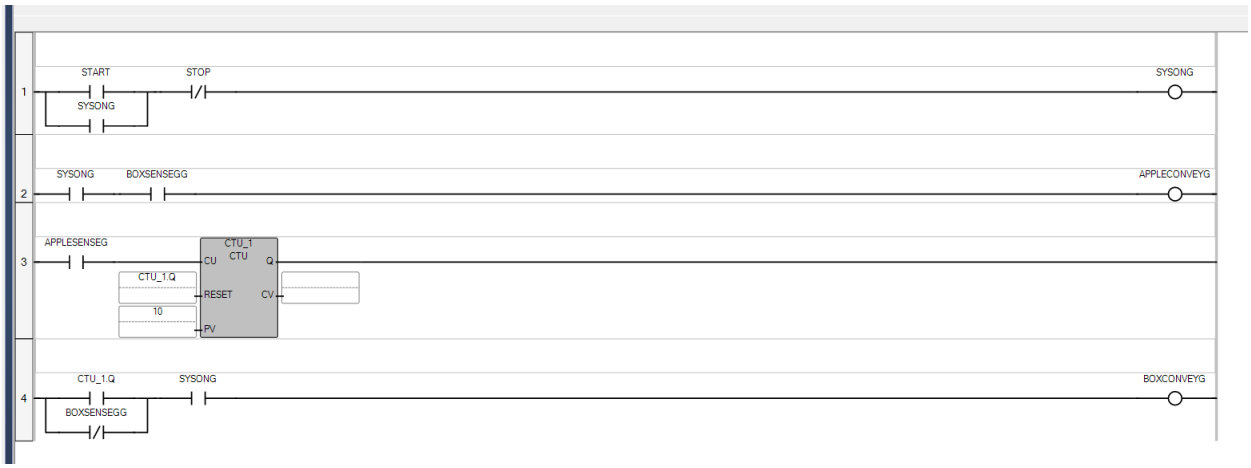
Apple Conveyor: Transports apples towards a packaging box.

Box Conveyor: Transports the filled box away once the target number of apples is reached.

Apple Sensor (APPLESENSEG): Detects each individual apple as it passes, triggering the counter.

Box Sensor (BOXSENSEGG): Detects the presence of an empty box ready to be filled.

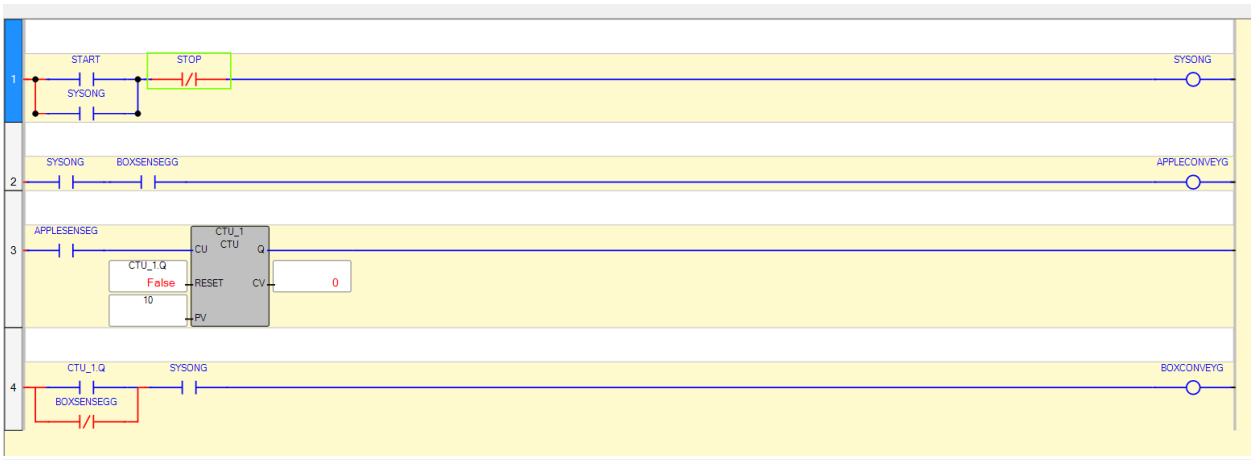
The process requires the system to start via a momentary push button, run the apple conveyor to fill a detected box, count the apples, and then activate the box conveyor to move the full box away once the target count is achieved.

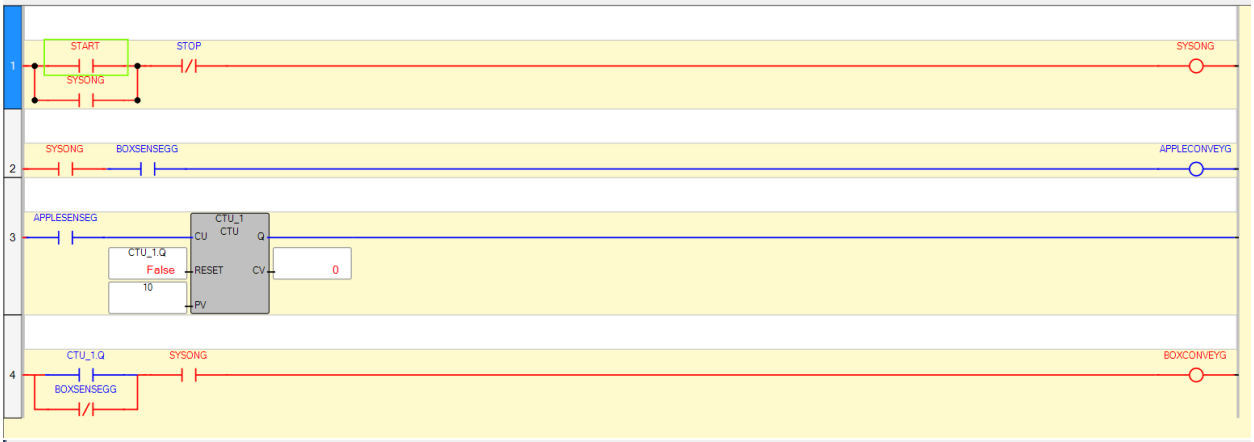


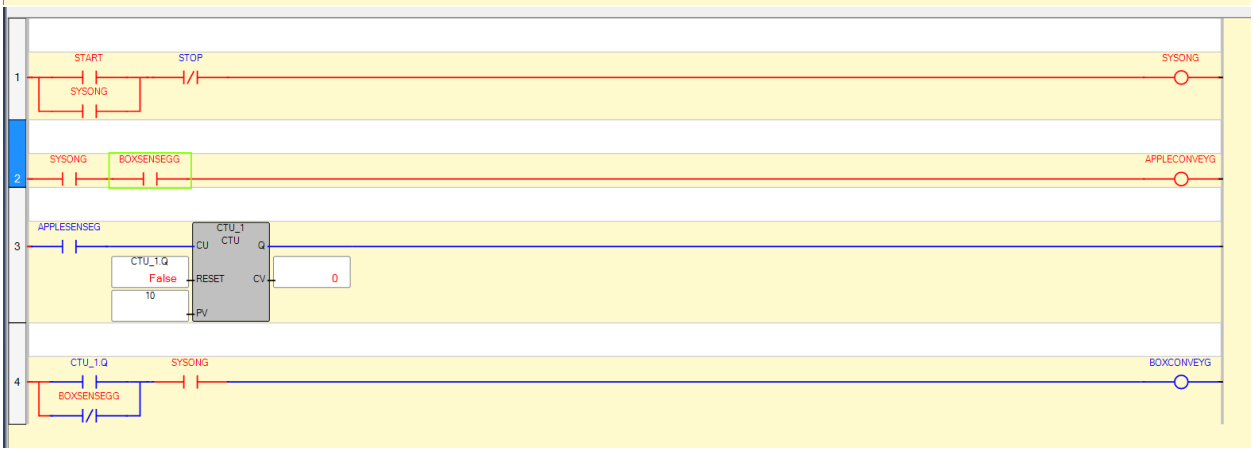
The Variables of the systems are:

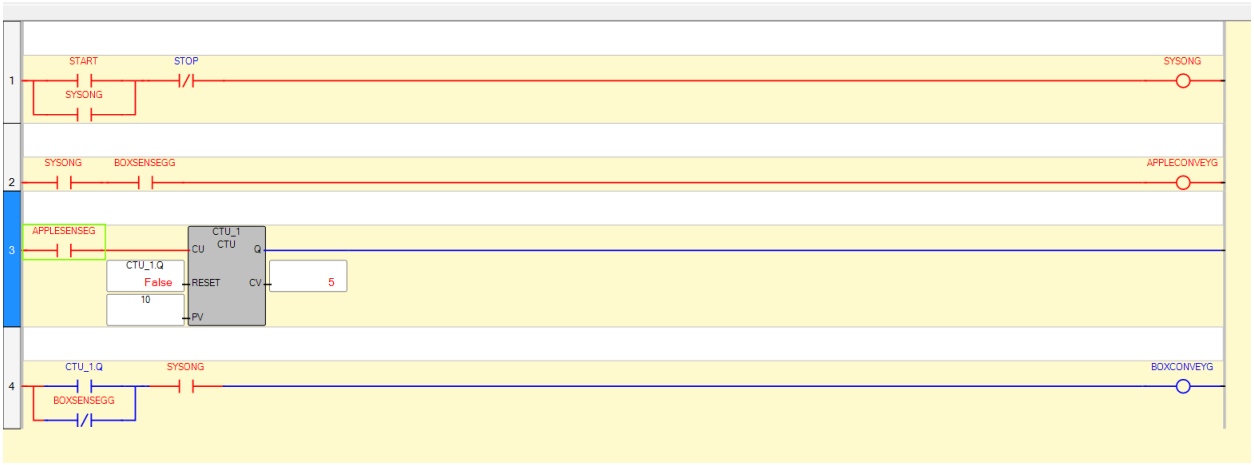
	Name	Alias	Data Type	Dimension	Project Val	Initial Value	Comment	String Size
	STOP		BOOL					
	SYSON		BOOL					
	BOXSENSE		BOOL					
	APPLECONVEY		BOOL					
	APPLESENSE		BOOL					
+	CTU_1		CTU		...	...		
	SYSON1		BOOL					
	BOXSENSEG		BOOL					
*								

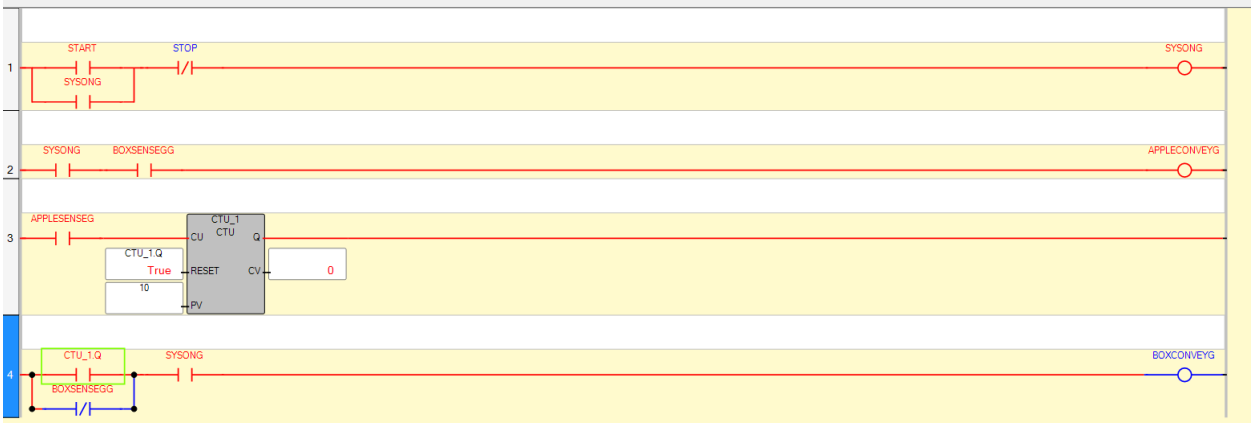
Output:

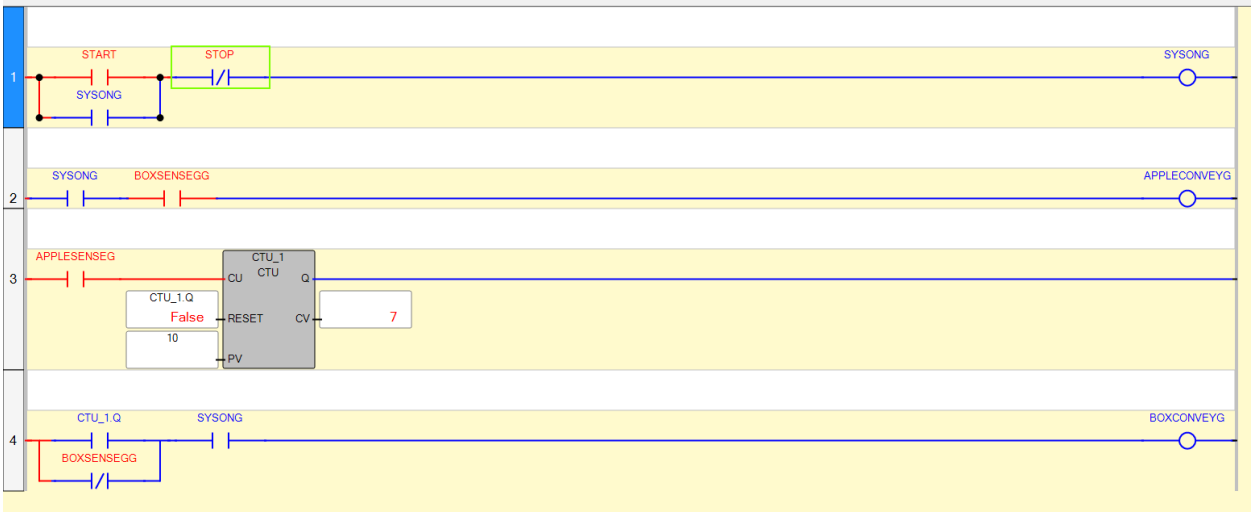

1
START
SYSONG
STOP
SYSONG
2
SYSONG
BOXSENSEGG
APPLECONVEYG
3
APPLESENSEG
CTU_1
CTU
CU
Q
CTU_1.Q
False
RESET
CV
0
10
PV
4
CTU_1.Q
SYSONG
BOXSENSEGG
BOXCONVEYG


1
START
SYSONG
STOP
SYSONG
2
SYSONG
BOXSENSEGG
APPLECONVEYG
3
APPLESENSEG
CTU_1
CTU
CU
Q
CTU_1.Q
False
RESET
CV
0
10
PV
4
CTU_1.Q
SYSONG
BOXSENSEGG
BOXCONVEYG


1
START
SYSONG
STOP
SYSONG
2
SYSONG
BOXSENSEGG
APPLECONVEYG
3
APPLESENSEG
CTU_1
CTU
CU
Q
CTU_1.Q
False
RESET
CV
0
10
PV
4
CTU_1.Q
SYSONG
BOXSENSEGG
BOXCONVEYG







Discussion:

The output shows the ladder logic in action. We first start the system using “START” push button. Then the “BOXSENSEGG” is on, it ensures that there is box to fill the apples. Now “APPLESENSEGG” sense the apple, then the counter counts the apple this increase the counter to +1. When it reaches 10, one box is filled then “BOXCONVEYER” is turned on. The whole system can be turned off using “STOP” push button.

Conclusion:

This lab successfully demonstrated the design and simulation of an industrial automated sorting and counting system using Rockwell CCW. The experiment reinforced the practical application of fundamental PLC programming concepts, specifically latching circuits and counter instructions.